

Exercise 13.2

1. A researcher collected data on number of deaths from Horse-Ricks in Russian Army crops over to years. The table is as follows:

No. of death	0	1	2	3	4	5	6
Frequency	60	50	87	40	32	15	10

Find the relative frequency of the given data.

No. of Deaths (x)	Frequency (f)	Relative Frequency ($\frac{f}{294}$)
0	60	$\frac{60}{294} = \frac{30}{147}$
1	50	$\frac{50}{294} = \frac{25}{147}$
2	87	$\frac{87}{294} = \frac{29}{98}$
3	40	$\frac{40}{294} = \frac{20}{147}$
4	32	$\frac{32}{294} = \frac{16}{147}$
5	15	$\frac{15}{294} = \frac{5}{98}$
6	10	$\frac{10}{294} = \frac{5}{147}$
Total	$\Sigma f = 294$	

2. The frequency of defective products in 750 samples are shown in the following table.

No. of defectives per sample	0	1	2	3	4	5	6	7	8
No. of sample	120	140	94	85	105	50	40	66	50

Find the relative frequency for the given table.

No. of Defectives (z)	No. of Samples (f)	Relative Frequency ($\frac{f}{750}$)
0	120	$\frac{120}{750} = \frac{4}{25}$
1	140	$\frac{140}{750} = \frac{14}{75}$
2	94	$\frac{94}{750} = \frac{47}{375}$
3	85	$\frac{85}{750} = \frac{17}{150}$
4	105	$\frac{105}{750} = \frac{7}{50}$
5	50	$\frac{50}{750} = \frac{1}{15}$
6	40	$\frac{40}{750} = \frac{4}{75}$
7	66	$\frac{66}{750} = \frac{11}{125}$
8	50	$\frac{50}{750} = \frac{1}{15}$
Total	$\Sigma f = 750$	

3. A quiz competition on general knowledge is conducted. The number of corrected answers out of 5 questions for 100 sets of questions is given below.

X	0	1	2	3	4	5
f	10	23	15	25	18	9

Find the relative frequency.

X	f	Relative Frequency $\left(\frac{f}{100}\right)$
0	10	$\frac{10}{100} = \frac{1}{10}$
1	23	$\frac{23}{100} = \frac{23}{100}$
2	15	$\frac{15}{100} = \frac{3}{20}$
3	25	$\frac{25}{100} = \frac{1}{4}$
4	18	$\frac{18}{100} = \frac{9}{50}$
5	9	$\frac{9}{100} = \frac{9}{100}$
Total	$\Sigma f = 100$	

4. A survey was conducted from the 108 students of a class and asked about their favourite food. The responses are as under:

Name of food item	Biryani	Fresh Juice	Chicken	Bar. B.Q	Sweets
No. of students	40	07	21	15	25

$$\text{Total students} = 40 + 7 + 21 + 15 + 25 = 108$$

(i) how many percentages of students like biryani?

$$\text{Percentage} = \frac{\text{No. of students for Biryani}}{\text{Total students}} \times 100$$

$$\text{Percentage} = \frac{40}{108} \times 100$$

$$\text{Percentage} = 37.04\%$$

(ii) how many percentages of students like chicken?

$$\text{Percentage} = \frac{\text{No. of students for Chicken}}{\text{Total students}} \times 100$$

$$\text{Percentage} = \frac{21}{108} \times 100$$

$$\text{Percentage} = 19.44\%$$

(iii) which food is the least like by the students?

From the table, **Fresh Juice** = 7 students is the lowest.

(iv) which food is the most prefer by the students?

From the table, **Biryani** = 40 students is the highest.

5. In 500 trials of a thrown of two dice, what is expected frequency that the sum will be greater than 8?

When two dice are thrown, the sample space will be:

$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$

$$n(S) = 36$$

Let A be the event getting the sum greater than 8.

$$A = \{(3,6), (4,5), (5,4), (6,3), (4,6), (5,5), (6,4), (5,6), (6,5), (6,6)\}$$

$$n(A) = 10$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{10}{36}$$

$$P(A) = \frac{5}{18}$$

Now by using formula

Expected frequency = Total number of trials $\times$ Probability of the event

$$E = N \times P(A)$$

$$E = 500 \times \frac{5}{18}$$

$$E = 138.9$$

6. What is the expectation of a person who is to get Rs. 120 if he obtains at least 2 heads in single toss of three coins?

When 3 coins are tossed, the sample space will be:

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

$$n(S) = 8$$

Let A the event of getting at least 2 heads in single toss

$$A = \{HHT, HTH, THH, HHH\}$$

$$n(A) = 4$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{4}{8}$$

$$P(A) = \frac{1}{2}$$

Now by using formula

Expected frequency = Total number of trials $\times$ Probability of the event

$$E = N \times P(A)$$

$$E = 120 \times \frac{1}{2}$$

$$E = 60$$

7. Find the expected frequencies of the given data:

x	0	1	2	3	4	5	6
$P(x)$	0.11	0.21	0.17	0.18	0.09	0.17	0.07

if the experiment is repeated 200 times.

x	$P(x)$	$E = N \times P(x)$ $E = 200 \times P(x)$
0	0.11	$200 \times 0.11 = 22$
1	0.21	$200 \times 0.21 = 42$
2	0.17	$200 \times 0.17 = 34$
3	0.18	$200 \times 0.18 = 36$
4	0.09	$200 \times 0.09 = 18$
5	0.17	$200 \times 0.17 = 34$
6	0.07	$200 \times 0.07 = 14$
Total	1.00	200

8. The probability of getting 5 sixes while tossing six dice is $\frac{2}{5}$, the dice is rolled 200 times. How many times would you expect it to show 5 sixes?

$$\text{Probability of getting 5 sixes} = P(5 \text{ sixes}) = \frac{2}{5}$$

Now by using formula

Expected frequency = Total number of trials $\times$ Probability of the event

$$E = N \times P(5 \text{ sixes})$$

$$E = 200 \times \frac{2}{5}$$

$$\mathbf{E = 80 \text{ times}}$$

Muhammad Tayyab (GHS Christian Daska)